Question	Answer	Marks	Partial Marks
4(c)	130 nfww or 129.6 to 129.8	**5**	**B1** for $\angle ACD = 21°$ or $\angle CAD = 69°$ **Method 1** **M2** for $\cos 21 = \frac{12}{AC}$ oe or **M1** for $\angle ADC = 90$ soi **M1** for $\pi(their\ AC/2)^2$ OR **Method 2** **M2** for $\frac{12}{sin138} = \frac{r}{sin21}$ oe or **M1** for $\angle COD = 138$ soi **M1** for $\pi(their\ r)^2$ OR **Method 3** **M2** for $\cos 21 = \frac{6}{OC}$ oe or **M1** for $\angle CXO = 90$ soi where X is the point where the perpendicular from O meets the chord CD **M1** for $\pi(their\ OC)^2$
4(d)	78.4 or 78.37 to 78.41	**3**	**M2** for $\frac{x}{360} \times 2 \times \pi \times 9.5 + 2 \times 9.5 = 4 \times 8$ oe or **M1** for $\frac{x}{360} \times 2 \times \pi \times 9.5$ After **M0, SC1** for $9.5x + 19 = 32$ oe
5(a)(i)	2.7 to 2.8	**1**	

Question	Answer	Marks	Partial Marks
5(a)(ii)	tangent ruled at $x = -2$	**B1**	
	6 to 10	**2**	**dep on B1** or a close attempt at tangent at $x = -2$ or **M1** for rise/run for *their* tangent, or close attempt, at any point Must see correct or implied calculation from a drawn tangent After **M0, SC1** for gradient of tangent (or close attempt) in range embedded in $y = mx + c$
5(a)(iii)	$y = 2x - 2$ ruled and $x = -2.9$ to -2.8 cao	**3**	**B2** for correct ruled line or **B1** for short line or for freehand line or broken line or ruled line with gradient 2 or with y-intercept at -2 (but not $y = -2$)
5(b)	A (4, 17) B (–1.5, 0.5)	**5**	**B4** for (–1.5, 0.5) and (4, 17), or for $x = 4$ and $x = -1.5$ OR **B3** for A(4, 17) or B(–1.5, 0.5) OR **M1** for $2x^2 - 2x - 7 = 3x + 5$ oe AND either **M2** for $(2x + 3)(x - 4)$ or **M1** for $2x(x - 4) + 3(x - 4)$ or $x(2x + 3) - 4(2x + 3)$ or $(2x + c)(x + d)$ where $cd = -12$ or $c + 2d = -5$ [c and d are integers] OR **M2** for $\frac{-their\ b \pm \sqrt{(their\ b)^2 - 4(their\ a)(their\ c)}}{2(their\ a)}$ or **M1** for $\sqrt{(their\ b)^2 - 4(their\ a)(their\ c)}$ or for $p = -their\ b$, $r = 2(their\ a)$ if in the form $\frac{p + \sqrt{q}}{r}$ or $\frac{p - \sqrt{q}}{r}$